

BT101

Introductory Biology

(Lecture 10)

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Introduction



Evolution of life:

- ☐ Origin of Life
- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Biomolecules or Biological Molecules



- ❑ **Biomolecules (also called biological molecules), are substances that are produced by the cells and living organisms.**
- ❑ **Biomolecules are of wide range of size and structures and perform a vast array of functions.**
- ❑ **The four major types of biomolecules are:**
 - 1) Nucleic acids**
 - 2) Proteins**
 - 3) Lipids**
 - 4) Carbohydrates**

Biomolecules or Biological Molecules



- ❑ **The four major types of biomolecules are carbohydrates, lipids, nucleic acids, and proteins.**
- ❑ Among biomolecules, nucleic acids, namely DNA and RNA, have the unique function of storing an organism's genetic code—the sequence of nucleotides that determines the amino acid sequence of proteins, which are of critical importance to life on Earth.
- ❑ There are 20 different amino acids that can occur within a protein; the order in which they occur plays a fundamental role in determining protein structure and function. Proteins themselves are major structural elements of cells. They also serve as transporters, moving nutrients and other molecules in and out of cells, and as enzymes and catalysts for the vast majority of chemical reactions that take place in living organisms. Proteins also form antibodies and hormones, and they influence gene activity.
- ❑ Likewise, carbohydrates, which are made up primarily of molecules containing atoms of carbon, hydrogen, and oxygen, are essential energy sources and structural components of all life, and they are among the most abundant biomolecules on Earth. They are built from four types of sugar units—monosaccharides, disaccharides, oligosaccharides, and polysaccharides.
- ❑ Lipids, another key biomolecule of living organisms, fulfill a variety of roles, including serving as a source of stored energy and acting as chemical messengers. They also form membranes, which separate cells from their environments and compartmentalize the cell interior, giving rise to organelles, such as the nucleus and the mitochondrion, in higher (more complex) organisms.
- ❑ All biomolecules share in common a fundamental relationship between structure and function, which is influenced by factors such as the environment in which a given biomolecule occurs.
- ❑ Lipids, for example, are hydrophobic (“water-fearing”); in water, many spontaneously arrange themselves in such a way that the hydrophobic ends of the molecules are protected from the water, while the hydrophilic ends are exposed to the water. This arrangement gives rise to lipid bilayers, or two layers of phospholipid molecules, which form the membranes of cells and organelles.
- ❑ In another example, DNA, which is a very long molecule—in humans, the combined length of all the DNA molecules in a single cell stretched end to end would be about 1.8 metres (6 feet), whereas the cell nucleus is about 6 μm (6×10^{-6} metre) in diameter—has a highly flexible helical structure that allows the molecule to become tightly coiled and looped. This structural feature plays a key role in enabling DNA to fit in the cell nucleus, where it carries out its function in coding genetic traits.

Nucleic Acids

- A nucleic acid is a polymeric macromolecule made up of repeated units of monomers called '**Nucleotides**'.
- Nucleic acids serve as repositories and transmitters of genetic information.
- There are two types of Nucleic acids
 - I. Deoxyribonucleic acid (DNA)
 - II. Ribonucleic acid (RNA).
- A **nucleotide** has three components:
 1. A Nitrogenous base (Purine and Pyrimidine)
 2. A Sugar (ribose and deoxyribose)
 3. A Phosphate group

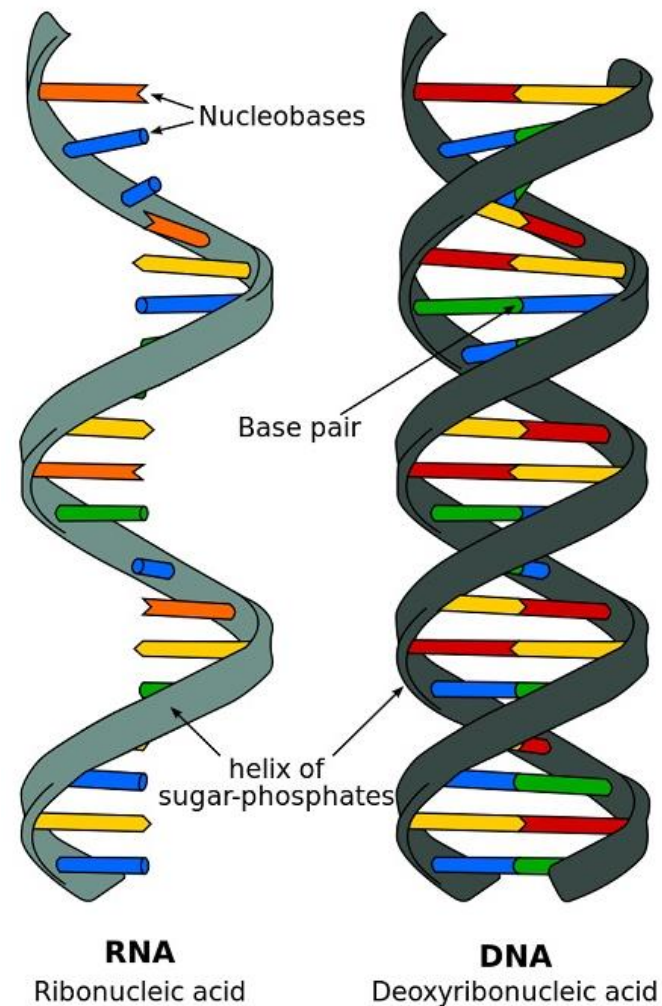
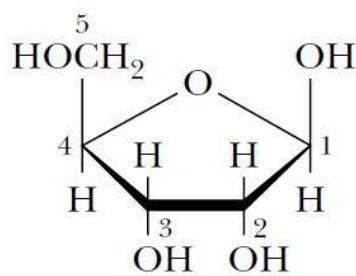


Fig. 9. General structure of RNA and DNA

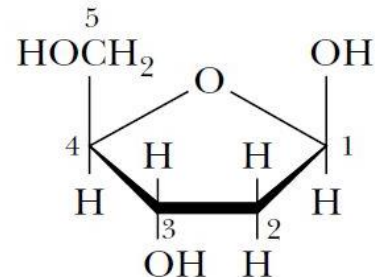
Nitrogenous bases and Sugars

- Nitrogenous bases found in nucleotides are aromatic heterocyclic compounds. The bases are of two types- **Purines** (Adenine and Guanine) and **Pyrimidines** (Cytosine, Thymine and Uracil).
- Five carbon containing monosaccharides (pentoses) are found in the nucleic acid structure. RNA contains **Ribose** while DNA contains **Deoxy-ribose**.



Furanose form of
D-Ribose

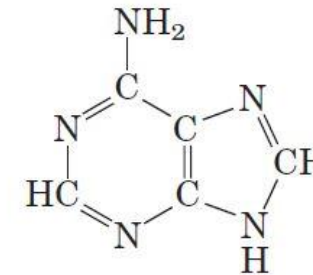
β -D-Ribofuranose



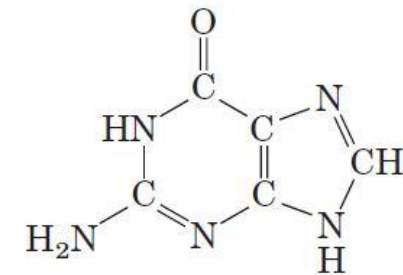
Furanose form of
D-2-Deoxyribose

β -D-2-Deoxyribofuranose

Fig. 10. Structures of Ribose and Deoxy-ribose 5 carbon sugars present in nucleic acids

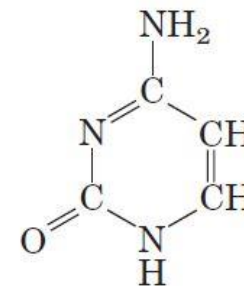


Adenine

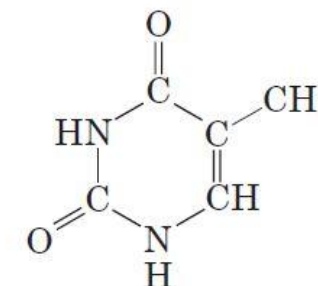


Guanine

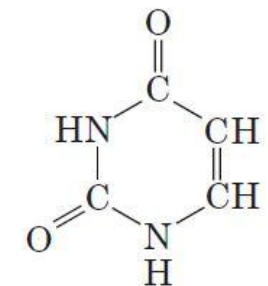
Purines



Cytosine



Thymine
(DNA)



Uracil
(RNA)

Pyrimidines

Fig. 11. Purines and Pyrimidines

Nucleotide

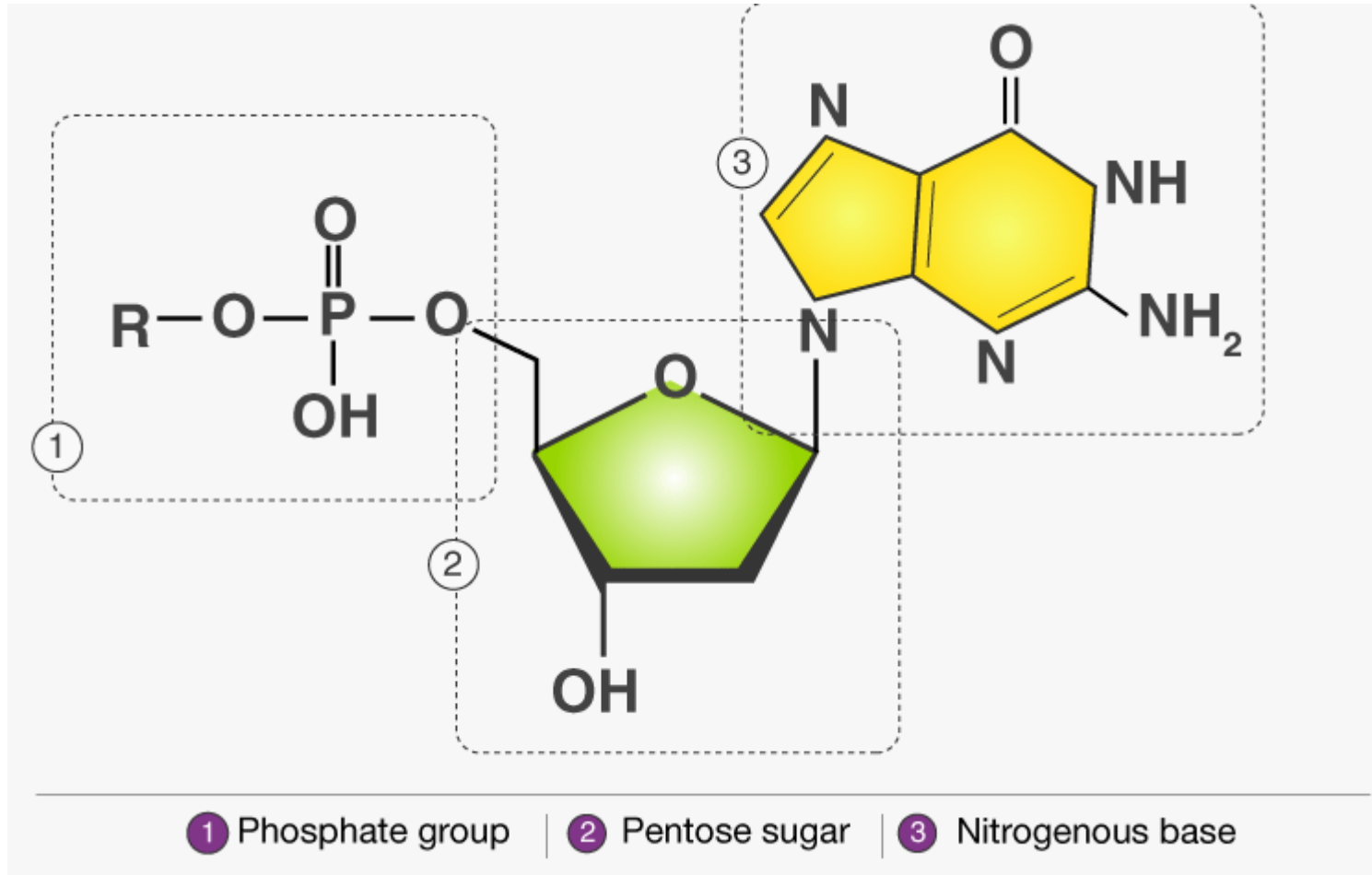
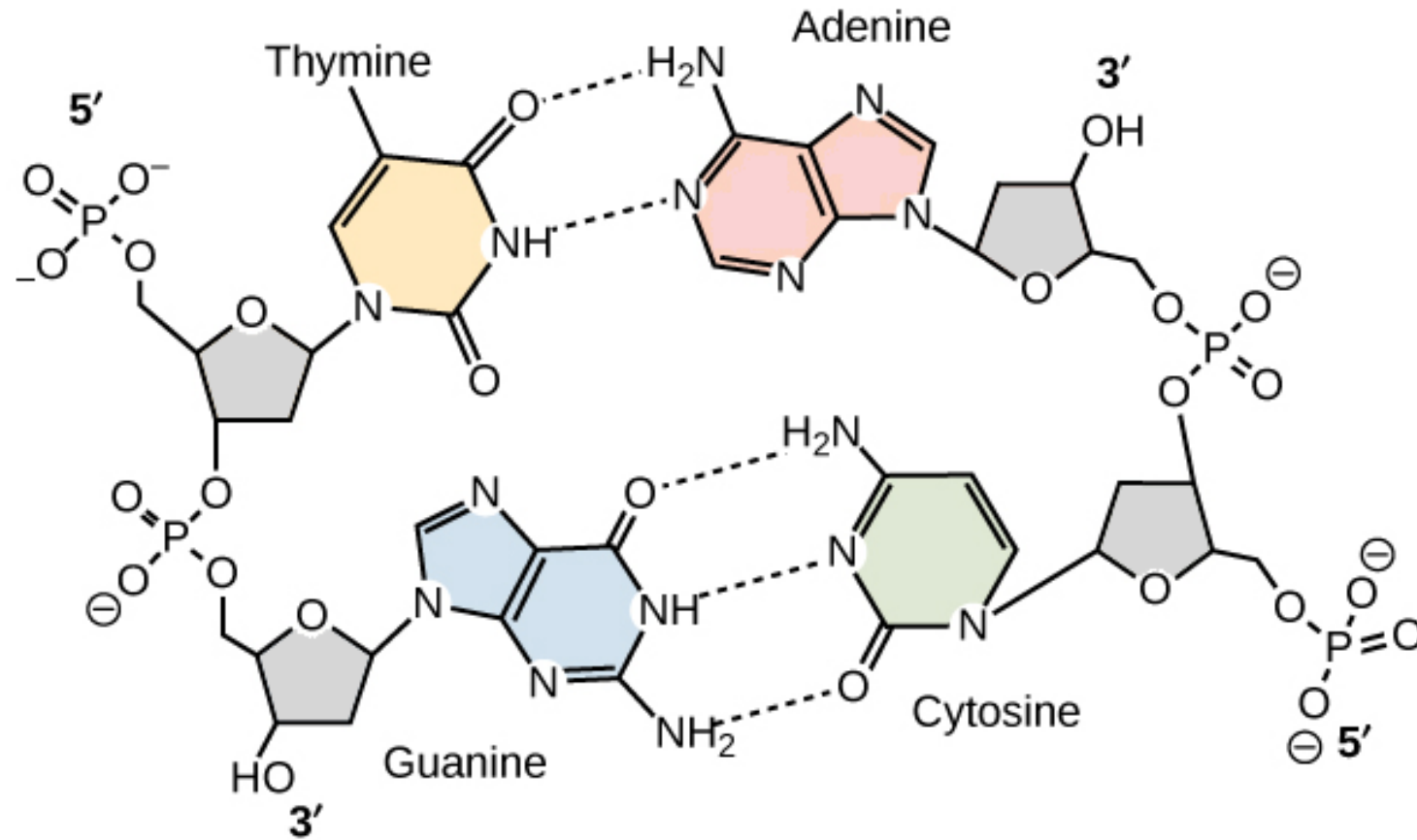
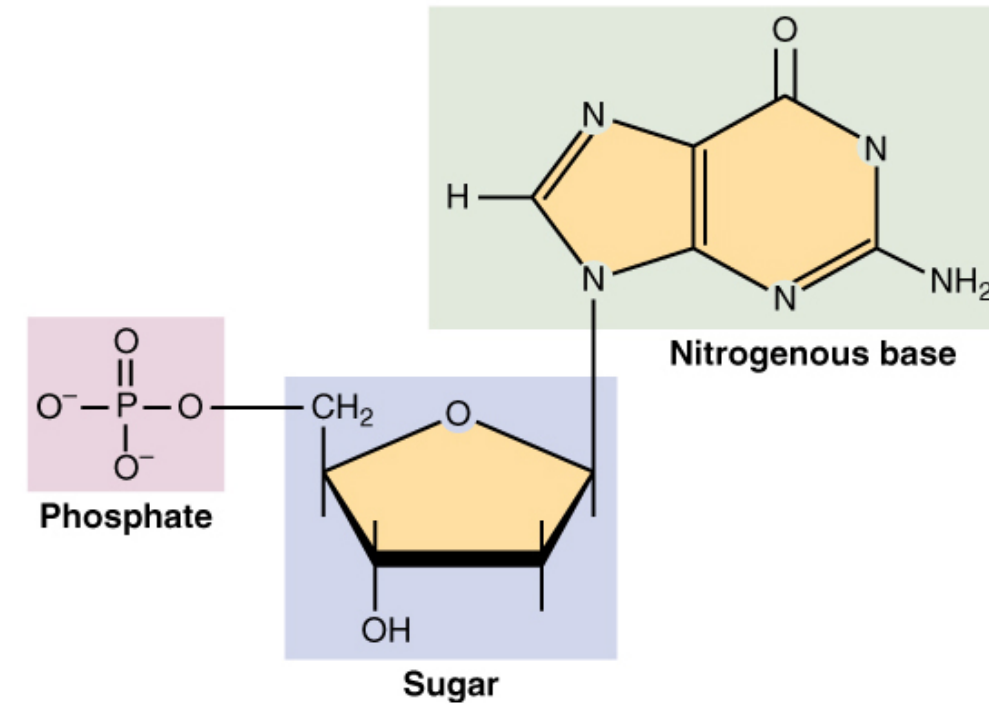
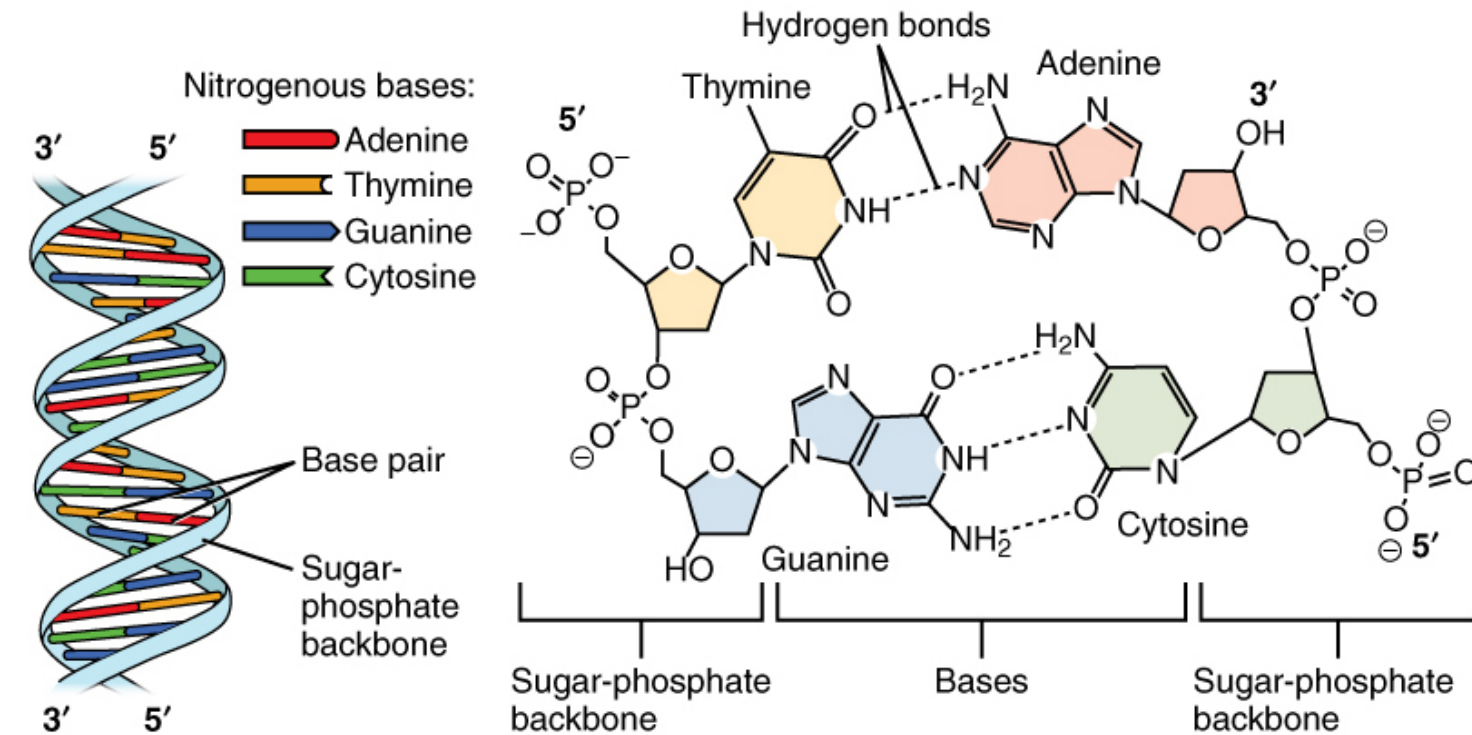


Fig. 12. A nucleotide (Adenosine mono phosphate, AMP) containing Adenine, Deoxyribose sugar and a Phosphohate group.

Nucleotide



Nucleotide



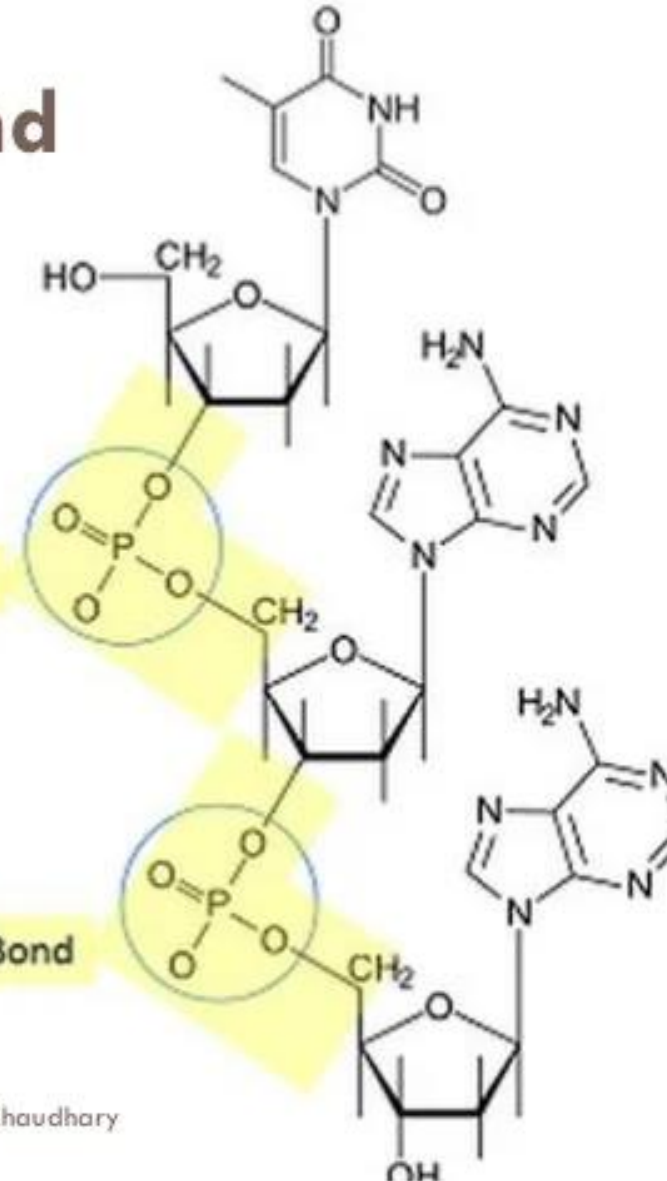
Phosphodiester bond

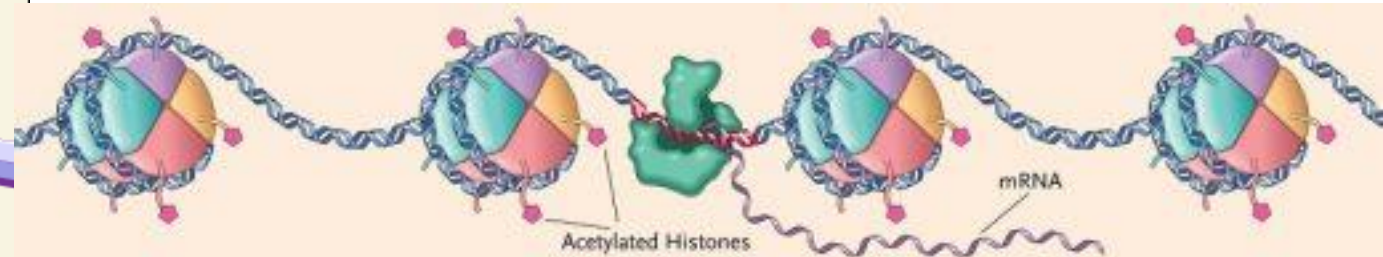
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Phosphodiester bonds in DNA or RNA are cleaved hydrolytically by chemicals or enzymes such as **nucleases: deoxyribonucleases** for DNA and **ribonucleases** for RNA.

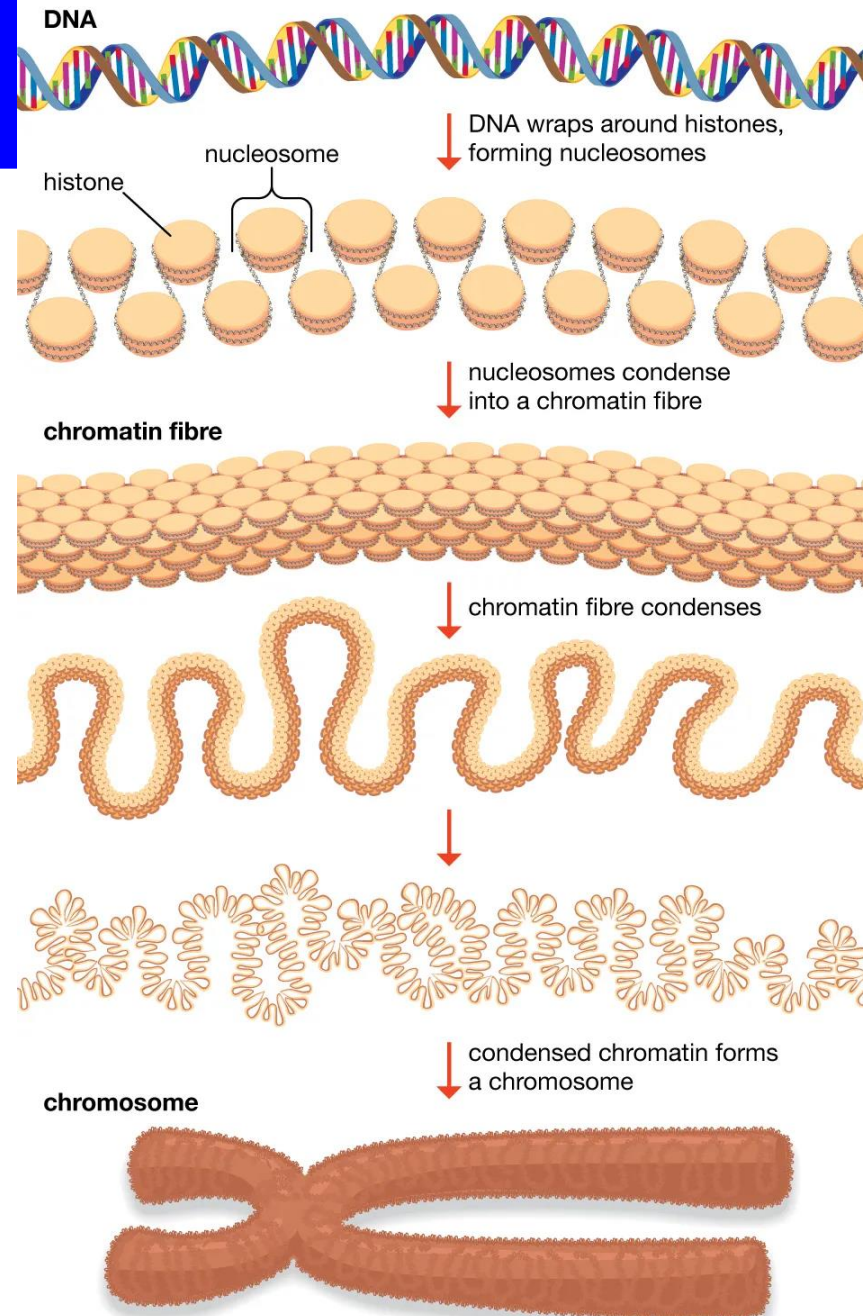
Phosphodiester Bond

Phosphodiester Bond





DNA packaging into chromatin and chromosome



Proteins

- Proteins are a diverse and abundant class of biomolecules, constituting more than 50% of the dry weight of cells. This diversity and abundance reflect the central role of proteins in virtually all aspects of cell structure and function.
- Proteins are Linear Polymers of Amino Acids.
- Amino acid, as the name signifies a molecule that contains an amino group and an acid group. It is an organic molecule that consists of a basic amino group, an acidic carboxyl group and a unique organic side chain.
- Proteins are unbranched polymers of amino acids linked from carboxyl group to amino group, through a type of amide linkage called **peptide bond**.

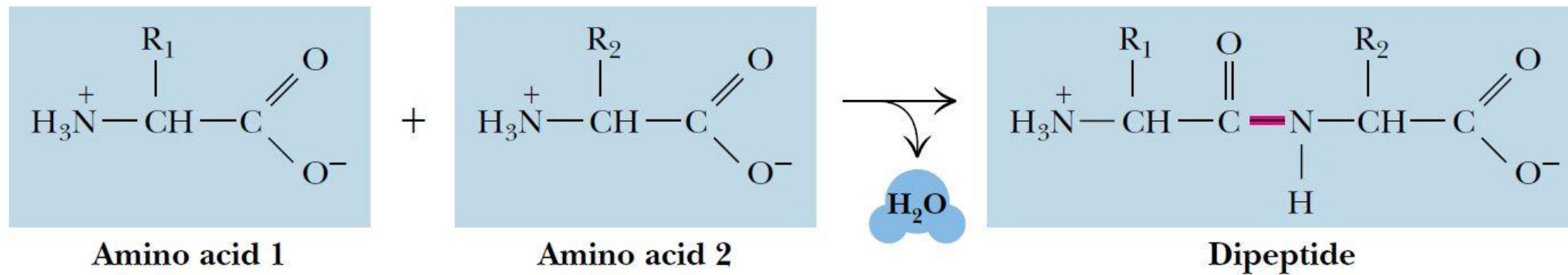


Fig. 8. Peptide bond formation is the creation of an amide bond between carboxyl group of one amino acid and the amino group of another amino acid. R_1 and R_2 represent, Side chain R groups of two different amino acids.

Amino acids, building blocks of Proteins

Amino acids are classified into two categories:

- i) **Essential amino acids**, which are not synthesized by the body and
- ii) **Non-essential amino acids**, which are synthesized by the body.

Table 1. Essential and Non-essential amino acids.

S. No.	Essential amino acids	S. No.	Non-essential amino acids
1	Histidine	01	Alanine
2	Isoleucine	02	Asparagine
3	Leucine	03	Aspartate
4	Lysine	04	Arginine
5	Methionine	05	Cysteine
6	Phenylalanine	06	Glutamate
7	Threonine	07	Glutamine
8	Tryptophan	08	Glycine
9	Valine	09	Proline
		10	Serine
		11	Tyrosine

Functional Classification of Proteins

Proteins are classified into various groups based on their function

- | | |
|----------------------------------|---|
| 1. Structural proteins | : Keratin of hair and nails, collagen of bone. |
| 2. Enzymes or catalytic proteins | : Hexokinase, pepsin (metabolism, digestion) |
| 3. Transport proteins | : Haemoglobin, serum albumin |
| 4. Hormonal proteins | : Insulin, growth hormone |
| 5. Contractile proteins | : Actin, myosin (muscle) |
| 6. Storage proteins | : Ovalbumin, glutelin (seeds) |
| 7. Genetic proteins | : Nucleoproteins (e.g. Histones) |
| 8. Defense proteins | : Immunoglobulins (antibodies), Snake venoms |
| 9. Receptor proteins | : Interferon receptor (e.g. cytokines) are signaling proteins released by host cells against virus. A virus-infected cell will release interferons causing nearby cells to heighten their anti-viral defenses. |

Lipids



- Lipids are a chemically diverse and heterogeneous group of compounds and it is difficult to define them precisely. The common and defining features of lipids are their low solubility in water and solubility in organic solvents. e.g. **fats, oils, waxes, phospholipids, sphingo-phospholipids, etc.,**
- **The biological functions of the lipids are as diverse as their chemistry.**
- Fats and oils are the principal stored forms of energy in many organisms.
- **Phospholipids and sterols are major structural elements of biological membranes.**
- Lipids play crucial roles as
 1. **Enzyme cofactors** : **Vitamin K (Phylloquinone)**
 2. Electron carriers : **Coenzyme Q (Mitochondria)**
 3. **Light-absorbing pigments** : **Chlorophyll**
 4. Hydrophobic anchors for proteins : **Glycosylphosphatidylinositol**
 5. **“Chaperones” to help membrane proteins fold** : **Phosphatidyl-ethanolamine**
 6. Emulsifying agents in the digestive tract : **Diglycerides**
 7. **Hormones** : **Testosterone**
 8. Intracellular messengers : **Diacylglycerol (DAG)**

Carbohydrates



- Carbohydrate is the most abundant biomolecule on earth. Each year, photosynthesis converts more than 100 billion metric tons of CO_2 and H_2O into cellulose and other plant products.
- Carbohydrate polymers (also called glycans) serve as structural and protective elements in the cell walls of bacteria, plants and in the connective tissues of animals.
- Complex carbohydrate polymers, that are covalently attached to the proteins or lipids act as signal molecules and are called **Glycoconjugates**.
- Carbohydrates are polyhydroxy aldehydes or ketones.
- There are three major size classes of carbohydrates: monosaccharides, oligosaccharides, and polysaccharides (the word “Saccharide” is derived from Greek *sakcharon*, meaning “sugar”).

Alteration of saccharon, from Ancient Greek, *via* Pali from Sanskrit शर्करा (*śárkarā*).

When the Greeks introduced from Asia Minor, it became in ancient Greece as Sakchar or Sakcharon.

Romans who subsequently took the word from the Greeks wrote it as Saccharum or Saccharon.

Carbohydrates

Carbohydrates

Monosaccharide

Glucose
Fructose
Galactose



Single sugar
molecule

Disaccharide

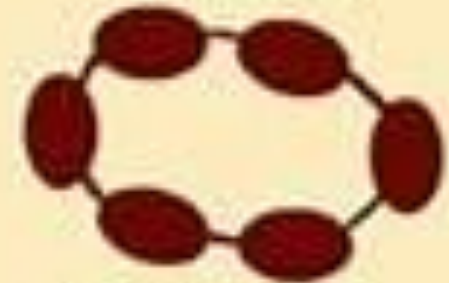
Maltose
Sucrose
Lactose



Two sugar
molecules linked

Polysaccharide

Starch
Glycogen
Cellulose



Many sugar
molecules linked

Mono-, 1.2 Di- and Oligo-saccharides

- **Monosaccharides**, or simple sugars, consist of a single polyhydroxy aldehyde or ketone unit. The two families of monosaccharides are aldoses and ketoses (Fig. 1a).
- The Most abundant monosaccharide in nature is the six-carbon reducing sugar, **D-Glucose**, also referred to as **Dextrose**.
- **Disaccharides** are made of two covalently linked monosaccharide units. The most abundant disaccharide is sucrose (from sugarcane), consisting of 6-carbon atom containing, D-glucose and D-fructose (Fig. 1b).
- **Oligosaccharides** are short chains of monosaccharide units, linked by glycosidic bonds.

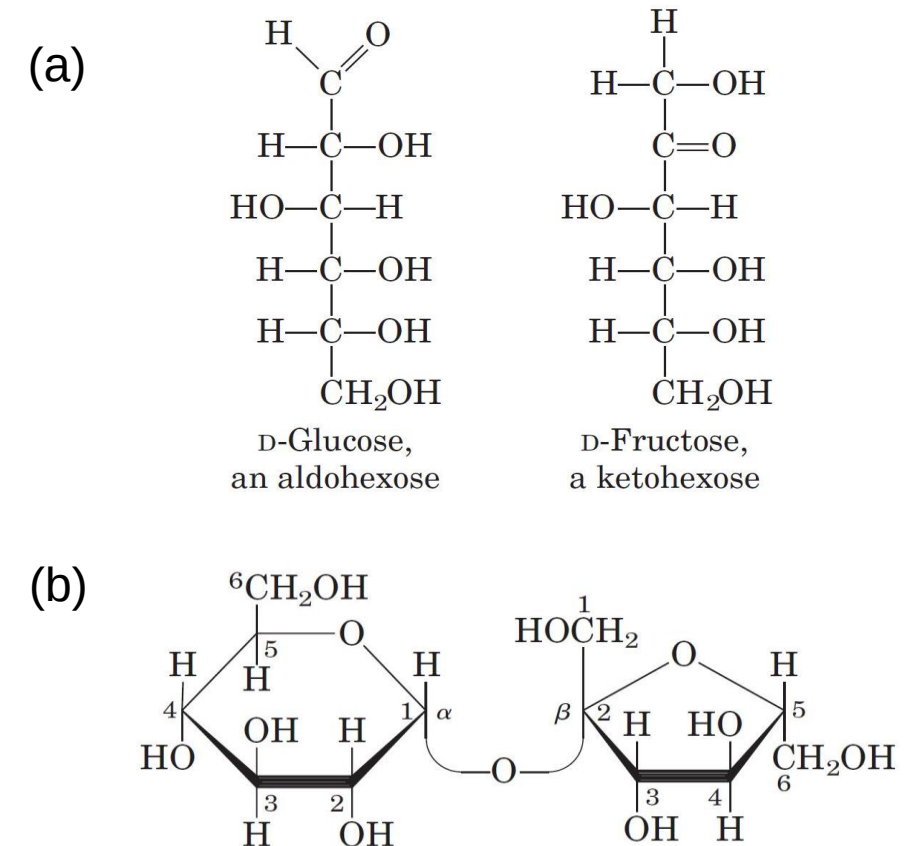


Fig. 1.

a) Representative monosaccharides.

D-Glucose an aldose and D-Fructose, a ketose.

b) Sucrose, a Disaccharide.

Polysaccharides (Cellulose, Starch and Glycogen)

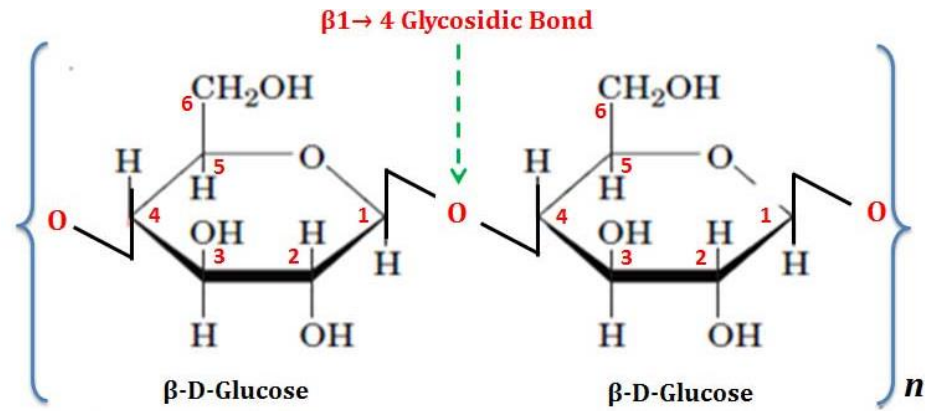


Fig. 3. Cellulose. Linear chain of D-glucose units.
Two D-glucose units joined by ($\beta 1 \rightarrow 4$) linkage.

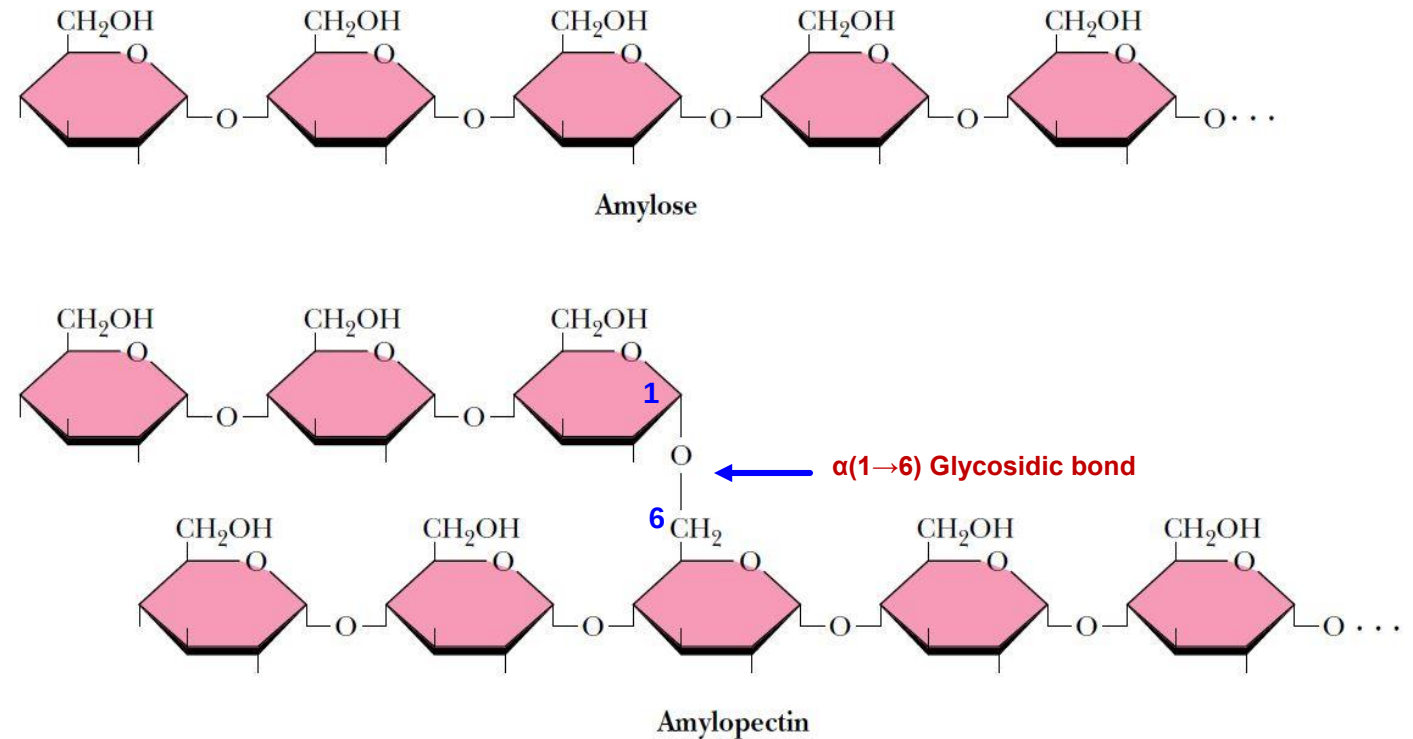


Fig. 4. Starch (Amylose and Amylopectin) and Glycogen (highly branched amylopectin).

Starch contains two types of glucose polymers, **Amylose** (Linear $\alpha(1 \rightarrow 4)$ linkages) and **Amylopectin** (Branched by $\alpha(1 \rightarrow 6)$ linkages), (less soluble in water and present in Plants as storage polysaccharide).

Glycogen has the same basic structure, but has more branching than **Amylopectin**. (water soluble and present in fungi and Animals as storage polysaccharide)

Important Links

Macromolecules

- ☐ <https://www.youtube.com/watch?v=FgfkNBZaVTI>
- ☐ <https://www.youtube.com/watch?v=YO244P1e9QM>
- ☐ <https://www.youtube.com/watch?v=GhABWQC3YDs> (Nucleic Acids)
- ☐ <https://www.youtube.com/watch?v=SeOrvA9ikW8> (Nucleic Acids)
- ☐ <https://www.youtube.com/watch?v=kMg517MHDJs> (Proteins)
- ☐ <https://www.youtube.com/watch?v=ebScOnAJdu0> (Lipids)
- ☐ <https://www.youtube.com/watch?v=rQyWJIn1HYE> (Carbohydrates)
- ☐ <https://www.youtube.com/watch?v=Mb2bJgEZtwE> (Replication)
- ☐ <https://www.youtube.com/watch?v=YlOql3PQwjo> (Transcription)

Ghee vs Butter

- ☐ <https://www.youtube.com/watch?v=DXPUWbJgmIE>